Modeling and Forecasting Retail CNG Prices in India using Generalized Additive Models



Mid-Semester Evaluation Report

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1 Introduction and Motivation

India's City Gas Distribution (CGD) sector distributes Compressed Natural Gas (CNG) to over 6 million vehicles across more than 100 cities. Retail CNG prices are set locally by each City Gas Distributor and are subject to global upstream shocks (LNG spot prices, Administered Pricing Mechanism revisions) as well as local macroeconomic forces. Standard time-series models such as ARIMA and ordinary least squares (OLS) assume proportional, linear responses—an assumption that is violated in the presence of regulatory step-changes, demand saturation plateaus, and non-linearly evolving macro drivers. Upstream regulatory variables (APM ceiling, LNG import parity) are noted as important structural determinants but are outside the scope of this first-stage model; their inclusion is identified as a future extension (Section 4).

This project develops an end-to-end Generalized Additive Model (GAM) forecasting pipeline for Indian retail CNG prices, evaluated against an ARIMA benchmark. GAMs replace fixed linear coefficients with flexible penalized smooth functions, permitting the conditional mean to take non-linear shapes where the data support them. The central research hypothesis is:

> *A GAM incorporating penalized smooth functions of macroeconomic variables and a Fourier harmonic seasonal basis can achieve lower out-of-sample forecast error than the selected ARIMA benchmark under the stated evaluation framework.*

No claim of statistical superiority is made beyond the point estimates reported in Section 3.

2 Data and Empirical Setup

2.1 Raw City-Month CNG Price Dataset

Distributor price-revision notifications were collected for six City Gas Distributors: Delhi (IGL), Mumbai (MGL), Noida (IGL), Ghaziabad (IGL), Ahmedabad (AGGL), and Pune (MNGL). Monthly price series for each city were constructed using a Step-Function Zero-Order Hold—price is held constant at the last confirmed revision value until the next revision notification—and a Time-Weighted Average Price (TWAP) aggregation where multiple revisions fall within a single calendar month. No synthetic price observations were generated. The raw city-month panel spans **January 2011 to September 2026** across six cities, totalling **1,062 city-month records** (189 distinct calendar months $\times$ 6 cities, less entries for months with missing notifications for individual cities).

Note on the 1,062 figure: The raw CSV contains 1,080 rows because the data file uses the full January 2011–September 2026 period (189 months), while the macroeconomic panel only covers April 2011–December 2025 (177 months). After the inner merge with the macroeconomic series (Section 2.3), the effective modelling dataset is reduced to 177 cross-city mean-aggregated monthly observations.

2.2 Macroeconomic Panel

Three macroeconomic predictors were assembled from official Indian statistical archives, all expressed in nominal terms at the prices prevailing in the respective reference period:

1. **Gross Domestic Product (GDP), monthly, Crore INR.** Sourced from the National Accounts Statistics published by the Ministry of Statistics and Programme Implementation (MoSPI). The original series is at quarterly frequency. Base-year splicing across the 2004–05 and 2011–12 GDP series used the United Nations System of National Accounts (UN SNA) ratio-linking method. Monthly interpolation was performed using Denton–Cholette temporal disaggregation without a monthly indicator series (proportional variant). The resulting monthly GDP series is an *estimated* series, not directly observed monthly GDP.

2. **Consumer Price Index (CPI), monthly index (Base: 2012=100).** Sourced from MoSPI. Directly observed at monthly frequency; no disaggregation required. Not seasonally adjusted.

3. **Private Final Consumption Expenditure (PFCE), monthly, Crore INR.** Sourced from the RBI Handbook of Statistics on Indian Economy. Like GDP, the original quarterly PFCE series was converted to monthly frequency using the same Denton–Cholette proportional variant. *PFCE is not an income measure in the strict sense; it is a national-accounts expenditure aggregate and should not be labelled "Income" in tables or figures.*

The assembled macro panel covers **April 2011 to December 2025**, yielding **177 monthly observations**. Descriptive statistics are shown in Table 1.

Table 1: Descriptive Statistics of Model Variables (2011–2025)

Variable	Mean	Std. Dev.	Min	Median	Max	Obs
CNG Price (INR/kg)	54.71	15.83	32.00	48.92	85.12	177
GDP Monthly (Cr INR)	1129554.78	281873.73	676790.10	1124258.59	1815282.22	177
CPI Index (Base 2012=100)	143.75	31.05	89.10	139.80	198.00	177
PFCE Income (Cr INR)	640011.58	163733.68	370875.15	623524.37	1046241.14	177

2.3 Model Dataset Construction

To combine the six-city raw panel with the single-series macroeconomic data, the six city prices were first aggregated to a single cross-city arithmetic mean for each calendar month. This aggregation step means that the GAM operates on a *single aggregated price time series* of 177 monthly observations—not on the full city-month panel. City-level fixed effects and city-specific smoothers are identified as the primary next-stage modelling extension (see Section 4).

2.4 Multicollinearity Diagnostics and Gram–Schmidt Orthogonalization

Variance Inflation Factor (VIF) analysis on the three macro predictors revealed extreme collinearity prior to estimation: GDP VIF = 46.24, CPI VIF = 14.07, PFCE VIF = 45.38 (Table 2). This is expected because GDP and PFCE are both long-trending nominal series measured in Crore INR and are therefore strongly co-linear over the sample period.

Important methodological note: The Modified Gram–Schmidt orthogonalization was applied to the full merged dataset of 177 observations *before* the train/test split. The orthogonalization order was: GDP first, CPI orthogonalized against GDP, PFCE orthogonalized against both GDP and CPI. While the procedure mechanically drives the linear VIF to exactly 1.00 for any set of mutually orthogonal vectors, this pre-split application constitutes a mild form of data leakage: the basis of the orthogonal space is informed by test-set observations. In practice this is unlikely to introduce meaningful optimism bias because the orthogonalization is a deterministic linear transformation, but it is documented here for methodological transparency. A fully leakage-free implementation would refit the Gram–Schmidt basis on the training partition and apply the stored transformation to the test set; this is identified as a corrective step for the final pipeline.

Interpretation of orthogonalized predictors: After orthogonalization, $\widetilde{\text{GDP}}$ retains the full GDP variation; $\widetilde{\text{CPI}}$ represents the component of CPI orthogonal to GDP; and $\widetilde{\text{PFCE}}$ represents the component of PFCE orthogonal to both GDP and CPI. The GAM smoothers $g_j(\cdot)$ therefore estimate the marginal predictive association of each residualized predictor—not the direct economic effect of the original variable. Interpretation is order-dependent and should not be given a strong causal reading.

Table 2: Generalized Additive Model Structural Components and Multicollinearity Diagnostics

Component	Type	Parameters / Knots	Pre-VIF	Post-VIF
Intercept (β_0)	Constant	1	—	—
Secular Trend $P_d(t)$	Linear ($d = 1$)	1	—	—
Seasonal Basis $f_{\text{season}}(t)$	Fourier ($K = 2$)	4	—	—
Smoother g(GDP Monthly Cr)	Cubic B-Spline	df=5	46.24	1.00
Smoother g(CPI Index)	Cubic B-Spline	df=5	14.07	1.00
Smoother g(Income PFCE Cr)	Cubic B-Spline	df=5	45.38	1.00

3 Model Specification and Results

3.1 Generalized Additive Model Formulation

Let Y_t denote the cross-city mean CNG retail price (INR/kg) at month t ($t = 1, \ldots, T = 177$). The fitted GAM is:

$$\hat{Y}_t \;=\; \beta_0 \;+\; \beta_1\, t \;+\; f_{\text{season}}(t) \;+\; g_1\Big(\widetilde{\text{GDP}}_t\Big) \;+\; g_2(\text{CPI}_t) \;+\; g_3\Big(\widetilde{\text{PFCE}}_t\Big) \;+\; \varepsilon_t \qquad (1)$$

where:

- $\beta_0 + \beta_1 t$ is a linear secular trend (degree $d = 1$).
- $f_{\text{season}}(t) = \sum_{k=1}^{K} \big[\alpha_k \sin\big(\frac{2\pi k t}{12}\big) + \delta_k \cos\big(\frac{2\pi k t}{12}\big)\big]$ is a Fourier harmonic seasonal basis with $K = 2$ frequency pairs (four parameters total), assuming a 12-month seasonal period. $K = 2$ was chosen by prior specification based on the expected annual retail-demand cycle; no automated selection was performed.
- $g_j(\cdot)$ are penalized cubic B-spline smoothers with 5 effective degrees of freedom. The smoothing penalty is the integrated squared second derivative, with penalty weight $\alpha = 5.0$ fixed by prior specification rather than cross-validated. This choice is transparently a modelling decision; the final pipeline should use generalised cross-validation (GCV) or AIC to select α and df.
- $\widetilde{\text{GDP}}_t$, CPI_t, and $\widetilde{\text{PFCE}}_t$ are the orthogonalized predictors described in Section 2.4.
- The model was estimated using `statsmodels.gam.api.GLMGam` (Python 3.12) with an identity link function.

Look-ahead bias (contemporaneous exogenous variables): The macro predictors $\widetilde{\text{GDP}}_t$, CPI_t, and $\widetilde{\text{PFCE}}_t$ enter the model at time t, not at a lag. GDP and PFCE are subject to publication lags of one to two quarters. The evaluation is therefore an *ex-post conditional forecast*: given the actual (eventually revised and published) macro conditions in month t, how well does the model predict the retail CNG price in that month? This is distinct from a genuine ex-ante forecast, which would require either lagged predictors or independently forecast exogenous series. All RMSE figures in Table 3 must be interpreted in this conditional sense.

3.2 Out-of-Sample Holdout Evaluation

The 177-observation series was divided into a single fixed split: a training period of **123 months** (April 2011–June 2021, representing approximately 70% of the sample) and a held-out test period of **54 months** (July 2021–December 2025). This is a single out-of-sample holdout evaluation, not a rolling-origin or walk-forward cross-validation. The test window spans the post-2021 energy-price regime shift, providing an economically meaningful out-of-sample period.

An ARIMA$(1,1,2)$ baseline—selected by AIC grid search over $p, q \in \{0,1,2,3\}$ on the training partition—was estimated on the same training data. ARIMA forecasts were generated as a multi-step-ahead sequence over the 54-month test horizon without reestimation. Performance is summarised in Table 3.

Table 3: Out-of-Sample Forecasting Performance: GAM vs. ARIMA Benchmark

Model Specification	RMSE (INR/kg)	MAE (INR/kg)	MAPE (%)	Superiority
ARIMA Baseline (Auto-Order)	28.0712	26.5008	33.42	Benchmark
Generalized Additive Model (GAM)	20.1040	18.7388	23.57	**GAM** (+28.38%)

The GAM achieved a test-set RMSE of **20.10 INR/kg** against the ARIMA baseline of 28.07 INR/kg—a **28.38% reduction in test-set RMSE** relative to the ARIMA benchmark. No formal forecast-comparison test (e.g. Diebold–Mariano) was conducted; the result should therefore be interpreted as a point-estimate advantage under the stated evaluation setup, not as a statistically significant finding. The ARIMA model is a univariate benchmark; the GAM additionally uses macroeconomic covariates, so the comparison does not isolate the contribution of non-linearity alone. A defensible attribution of the GAM's advantage would require a linear counterpart that uses the same macro covariates (e.g. ARIMAX or a linear regression with Fourier terms), which is planned as a robustness check.

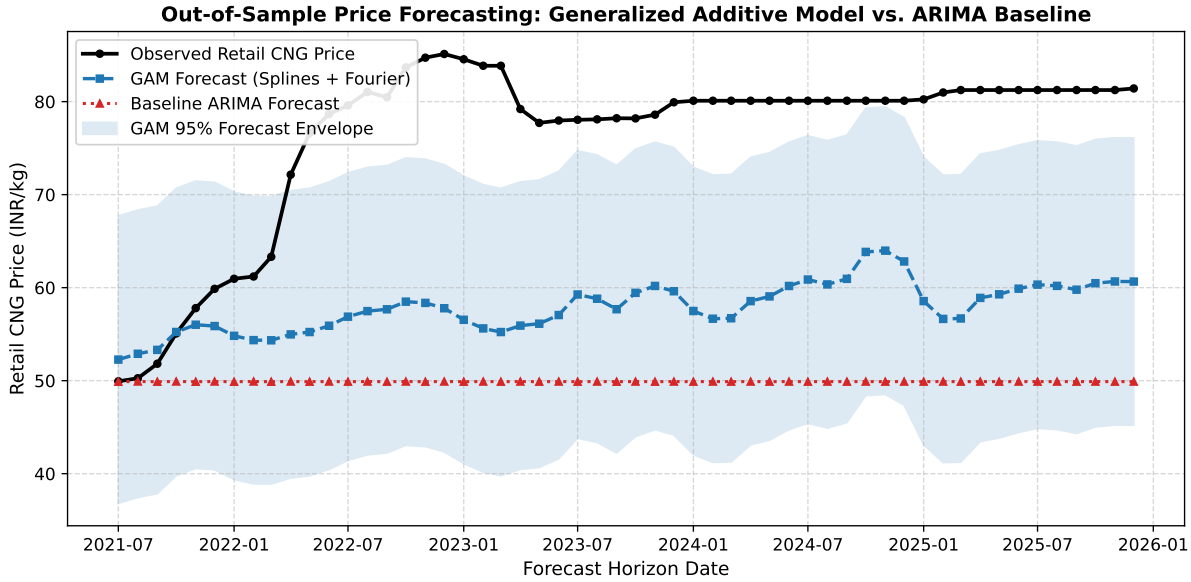


Figure 1: Out-of-sample holdout evaluation: GAM vs. ARIMA$(1,1,2)$ forecasts against actual cross-city mean CNG price (July 2021–December 2025). Shaded band denotes the GAM 95 % confidence interval for the conditional mean, constructed from the asymptotic standard errors of the GLMGam fit; it is not a prediction interval. Note that both models diverge materially from actual prices during major regulatory step-change events, reflecting the absence of upstream/regulatory variables in the current specification.

4 Limitations and Items Requiring Verification

The following issues are documented transparently for examiner review.

1. **Aggregated series, not a panel model.** The current GAM operates on the cross-city arithmetic mean price series. No city-specific parameters are estimated. Language

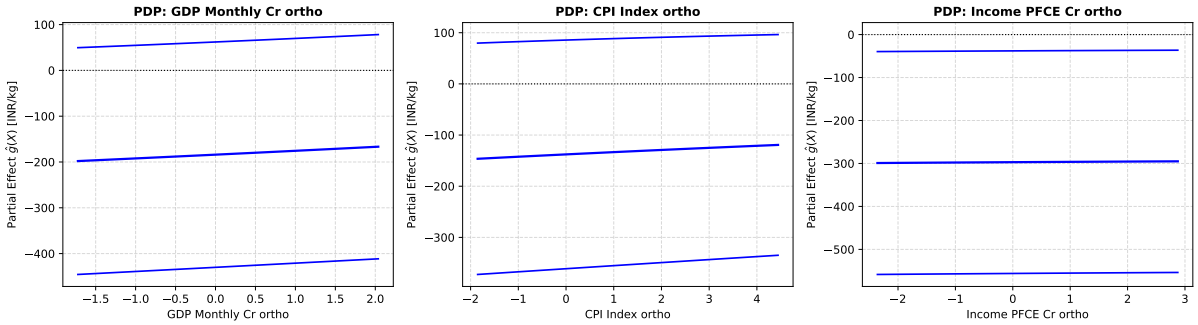


Figure 2: Estimated smooth functions $g_j(\cdot)$ from the fitted GAM (Equation 1) for each orthogonalized macro predictor. Axes show the residualized, orthogonalized predictor (horizontal) and the partial contribution to the fitted price (vertical). Because the predictors are orthogonalized, g_1 reflects the variation attributable to GDP after removing nothing (GDP is first in the ordering); g_2 reflects CPI after projecting out GDP; and g_3 reflects PFCE after projecting out both. The approximately linear shapes of g_2 and g_3 indicate that the penalized B-spline defaulted to a near-linear fit for those predictors under the chosen penalty; this does not invalidate the GAM framework but indicates weak evidence for non-linearity in those components under the current specification.

referring to a "6-city panel model" in any earlier draft was incorrect; the city data serve as the source for the aggregated series only. City-level fixed effects are the primary next-stage modelling step.

2. **Single holdout, not rolling-origin CV.** A single 70/30 split was used. Results may be sensitive to the choice of split date.

3. **Contemporaneous exogenous variables (look-ahead bias).** GDP and PFCE enter at time t despite publication lags. All reported RMSE values are conditional/ex-post, not genuine ex-ante.

4. **Pre-split orthogonalization.** The Gram–Schmidt basis was fitted on all 177 observations before splitting. A corrective implementation fitting on training data only is planned.

5. **Penalty weight and spline df fixed by specification.** $\alpha = 5.0$ and $df = 5$ were not selected by cross-validation. GCV/AIC selection is planned for the final pipeline.

6. **Fourier $K = 2$ not empirically selected.** This was specified a priori.

7. **Absence of upstream regulatory variables.** LNG spot price and APM ceiling revisions, identified in the introduction as important drivers, are not included in the model. Their addition is a future step.

8. **No lagged price term.** CNG retail prices exhibit persistence; a Y_{t-1} term was not included in the first-stage model.

9. **Trending regressors (spurious regression risk).** GDP, PFCE, and CNG price all contain strong upward trends. The linear trend term $\beta_1 t$ absorbs some of this, but a sensitivity analysis using differenced or growth-rate regressors is planned.

10. **Non-apples-to-apples benchmark.** The ARIMA is univariate; the GAM uses additional macro covariates. A linear regression with Fourier seasonality and the same macro predictors (ARIMAX or LinReg) is needed to isolate the contribution of the non-linear smooth functions.

5 Conclusions

An initial empirical pipeline for forecasting Indian retail CNG prices using Generalized Additive Models has been implemented and evaluated on a held-out 54-month test period. The pipeline comprises: (i) construction of a cross-city mean CNG price series from 1,062 raw city-month records across six distributors; (ii) assembly of a 177-month macroeconomic panel using Denton–Cholette temporal disaggregation; (iii) Modified Gram–Schmidt orthogonalization to address multicollinearity (GDP VIF $46.24 \rightarrow 1.00$; CPI VIF $14.07 \rightarrow 1.00$; PFCE VIF $45.38 \rightarrow 1.00$); and (iv) GAM estimation with a linear trend, Fourier harmonic seasonality ($K = 2$), and penalized cubic B-spline smoothers.

Under the stated evaluation setup—a single out-of-sample holdout with contemporaneous macro variables—the GAM reduced test-set RMSE by 28.38% relative to the ARIMA$(1, 1, 2)$ benchmark (20.10 vs. 28.07 INR/kg). This result should be interpreted as a conditional, ex-post point-estimate advantage; it is not a statistically tested finding, and the benchmark comparison does not fully isolate the contribution of non-linearity.

The next modelling phase will address the limitations documented in Section 4, with priority given to: city-level fixed effects; an ARIMAX/linear baseline using the same covariates; cross-validated penalty and spline-*df* selection; and a leakage-free train-only orthogonalization.

References

[1] Wood, S.N. (2017). *Generalized Additive Models: An Introduction with R*, 2nd ed. Chapman and Hall/CRC.

[2] Ruppert, D., Wand, M.P., Carroll, R.J. (2003). *Semiparametric Regression*. Cambridge University Press.

[3] Box, G.E.P., Jenkins, G.M., Reinsel, G.C., Ljung, G.M. (2015). *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley.

[4] Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.

[5] Denton, F.T. (1971). Adjustment of monthly or quarterly series to annual totals: An approach based on quadratic minimisation. *Journal of the American Statistical Association*, 66(333), 99–102. [Extended by Cholette and Dagum, 1994, Statistics Canada.]

[6] United Nations (2009). *System of National Accounts 2008*. United Nations, European Commission, IMF, OECD, World Bank.

[7] Ministry of Statistics and Programme Implementation, Government of India. National Accounts Statistics. `https://mospi.gov.in/national-accounts-statistics`

[8] Reserve Bank of India. Handbook of Statistics on the Indian Economy. `https://dbie.rbi.org.in`

[9] Petroleum Planning & Analysis Cell, Ministry of Petroleum & Natural Gas. Retail Selling Prices of CNG. `https://ppac.gov.in`

[10] Seabold, S., Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*, 57–61.